

PROJECT MANAGEMENT SYSTEM

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ABSTRACT

Project management plays a crucial role in ensuring the successful planning, execution, monitoring, and completion of organizational activities. Traditional project management solutions often require complex backend infrastructures, database systems, and extensive deployment procedures, which may not be suitable for lightweight academic or prototype environments. To address these challenges, this project presents ProjectPulse, a modern web-based Project Management System designed to provide comprehensive project and task tracking capabilities through a fully client-side architecture. ProjectPulse is developed using HTML5, CSS3, JavaScript, Bootstrap 5, Font Awesome, and Chart.js, delivering a responsive and interactive user experience. The system utilizes the browser's LocalStorage mechanism as its persistence layer, enabling storage and retrieval of project-related information without requiring server-side processing or database integration. The application supports essential project management functionalities, including project creation and management, task tracking, workflow visualization through a Kanban board, deadline monitoring via a calendar interface, team member management, activity logging, notification handling, and analytical reporting. The system features a modern glassmorphism-based user interface with responsive design principles, ensuring accessibility across different devices and screen sizes. Interactive dashboards provide real-time insights into project progress, task completion status, pending activities, workload distribution, and upcoming deadlines. Furthermore, the reporting module leverages graphical visualizations to assist users in evaluating project performance and making informed decisions. By combining simplicity, usability, and modern web technologies, ProjectPulse demonstrates an effective approach to implementing project management workflows without the complexity of traditional server-based architectures. The proposed system serves as a practical academic prototype while providing a foundation for future enhancements such as authentication mechanisms, cloud-based storage, real-time collaboration, automated notifications, and advanced reporting capabilities.

Keywords : Project Management System, Project Tracking, Task Management, Kanban Board, LocalStorage, Web Application, Dashboard Analytics, Workflow Management, Chart.js, Bootstrap 5, Client-Side Architecture, Responsive Web Design.

CHAPTER-1

INTRODUCTION

1.1 Introduction

This chapter presents the design and implementation of the proposed Project Management System, named ProjectPulse. The system is developed to provide a comprehensive and enterprise-oriented platform for managing project activities, including project planning, task allocation, progress monitoring, and performance reporting. The primary objective of ProjectPulse is to facilitate efficient project tracking through an intuitive and interactive web-based environment.

ProjectPulse is implemented as a lightweight client-side application that operates without requiring a dedicated backend server or database infrastructure. The system is specifically designed to demonstrate real-world project management workflows while maintaining simplicity, portability, and ease of deployment. To achieve persistent data storage, the application utilizes the browser's Local Storage mechanism, enabling project and task information to be retained across user sessions and page refreshes. The proposed system incorporates modern dashboard-based interfaces that enhance user experience and support effective decision-making. Key functionalities include project creation and management, task assignment and tracking, workflow visualization, progress analytics, and deadline monitoring. These features enable users to maintain visibility over project activities and assess overall project performance in real time.

To improve project execution and collaboration, ProjectPulse integrates multiple visualization components, including progress indicators, analytical charts, Kanban-based workflow management, and calendar-driven scheduling. The Kanban board facilitates task movement across different stages of completion, while analytical dashboards provide insights into project status and productivity metrics. Additionally, the calendar module assists users in monitoring important deadlines and upcoming milestones. By combining usability, visualization, and data persistence within a single web application, ProjectPulse serves as an effective demonstration of modern project management principles. The system highlights how contemporary web technologies can be utilized to develop scalable and user-friendly project tracking solutions without the complexity of traditional server-based architectures.

1.2 Proposed System

The proposed system, ProjectPulse, is a web-based Project Management System developed to provide an efficient and user-friendly environment for project planning, task management, progress monitoring, and performance analysis. The application is implemented using modern front-end technologies, including HTML5, CSS3, and JavaScript, to ensure responsiveness, maintainability, and cross-platform compatibility. The user interface is enhanced through the integration of Bootstrap 5, while Font Awesome is utilized for graphical icons and visual elements.

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Furthermore, Chart.js is employed to generate analytical reports and interactive data visualizations. The system is designed as a lightweight client-side application that operates independently of server-side infrastructure. By utilizing the browser's LocalStorage mechanism for data persistence, ProjectPulse provides a practical and cost-effective solution for managing project-related information without requiring database connectivity or backend services.

1.2.1 System Architecture

ProjectPulse follows a three-layer client-side architecture consisting of the Presentation Layer, Application Logic Layer, and Persistence Layer. This architectural design promotes modularity, maintainability, and efficient data management.

Presentation Layer

The Presentation Layer serves as the user interface of the application and consists of multiple functional modules, including Dashboard, Projects, Tasks, Kanban Board, Calendar, Reports, Team Management, and Settings. The interface is designed using HTML5 and Bootstrap 5, complemented by custom CSS styling based on a modern glassmorphism design approach. This layer facilitates user interaction and presents project information in an organized and visually appealing manner.

Application Logic Layer

The Application Logic Layer is implemented using JavaScript and is responsible for handling user interactions, data processing, event management, and dynamic content rendering. A centralized JavaScript module manages application workflows and coordinates communication between the user interface and the persistence layer. This layer ensures that all project operations are executed efficiently and consistently across different modules.

Persistence Layer

The Persistence Layer utilizes the browser's LocalStorage API to maintain application data across user sessions. All project-related information, including projects, tasks, team members, activities, notifications, and user preferences, is stored locally within the browser.

1.2.2 Data Management Using LocalStorage

Data persistence within ProjectPulse is achieved through a structured LocalStorage-based storage mechanism. Separate storage keys are maintained for each primary data entity, enabling organized data retrieval and update operations.

The system incorporates several data management functionalities, including:

- Retrieval of previously stored data during application initialization.
- Automatic generation of default datasets when no existing records are available.

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- Storage of modifications resulting from user actions such as creation, editing, deletion, archiving, and status updates.

The adoption of LocalStorage provides a lightweight and efficient solution for data management while maintaining the operational simplicity required for academic project implementation.

CHAPTER-2

IMPLEMENTATION & DETAILS

2.1 Introduction

This chapter presents the implementation methodology and technical details of the proposed Project Management System, namely ProjectPulse. The system is designed to provide a modern, interactive, and efficient platform for project planning, task management, workflow monitoring, and performance analysis. By integrating project tracking functionalities with intuitive visualization mechanisms, ProjectPulse enables users to effectively manage project activities, monitor task progress, and evaluate project performance in real time. Unlike conventional project management applications that depend on backend servers, database systems, and cloud-based services, ProjectPulse adopts a lightweight client-side architecture. The entire application is executed within the web browser, allowing it to function as a standalone static web application. This architectural approach eliminates the need for server-side deployment, reduces system complexity, and enhances portability while maintaining a rich and responsive user experience.

To ensure persistent data storage, the system utilizes the browser's Local Storage mechanism. All project-related information, including project records, task details, team member profiles, activity logs, notifications, and user preferences, is stored locally within the browser environment. This enables users to retain application data across sessions without requiring external database connectivity. The implementation of ProjectPulse is based on modern web technologies, including HTML5, CSS3, JavaScript, Bootstrap 5, Font Awesome, and Chart.js. These technologies collectively contribute to the development of a responsive, visually appealing, and feature-rich application capable of supporting enterprise-style project management workflows.

The primary objective of this chapter is to describe the design and implementation aspects of the proposed system, including its architectural framework, data management strategy, module organization, and user interface design. Furthermore, this chapter explains the implementation of key functionalities such as dashboard analytics, project and task management, Kanban-based workflow tracking, calendar scheduling, reporting mechanisms, and LocalStorage-based persistence. The discussion also highlights the techniques adopted to provide an intuitive and professional user experience while maintaining simplicity, scalability, and ease of deployment.

Through these implementation strategies, ProjectPulse demonstrates how modern front-end technologies can be effectively utilized to develop a comprehensive project management solution without relying on traditional server-side infrastructures.

The requirements for Enhancing End-to-End Encrypted Messaging: A Hybrid Classical–Post-Quantum Approach with Real-Time Multi-User Support come from the need for an user-friendly communication platform.

2.2 Explanation of Key Functions

The functionality of ProjectPulse is primarily implemented through a centralized JavaScript module that manages application logic, data processing, user interactions, and dynamic interface rendering. The system adopts a modular programming approach in which related operations are encapsulated within dedicated helper functions. This design improves code maintainability, reusability, and scalability while ensuring consistent behavior across all functional modules. The key functions responsible for data persistence, activity management, and user notifications are discussed in the following sections.

2.2.1 LocalStorage Data Access Functions

To provide persistent storage without relying on external databases, ProjectPulse utilizes the browser's LocalStorage API. A set of wrapper functions has been implemented to simplify data access operations and ensure consistency throughout the application.

Load Function

The `load(key, fallback)` function is responsible for retrieving data from LocalStorage. It reads JSON-formatted values associated with a specified storage key and converts them into JavaScript objects. If the requested data is unavailable, corrupted, or improperly formatted, the function returns a predefined fallback value. This mechanism prevents runtime errors and ensures reliable application initialization.

The primary responsibilities of this function include:

- Retrieving stored application data.
- Parsing serialized JSON objects.
- Handling missing or invalid data gracefully.
- Providing default values when necessary.

Save Function

The `save(key, value)` function manages the storage of application data within LocalStorage. Before storage, JavaScript objects are serialized into JSON format to ensure compatibility with browser storage mechanisms. This function is invoked whenever modifications occur within the system, such as project creation, task updates, configuration changes, or workflow transitions.

The major responsibilities of this function include:

- Serializing application data into JSON format.
- Updating LocalStorage records.
- Preserving user-generated information across browser sessions.
- Maintaining synchronization between the user interface and stored data.

Together, the Load and Save functions establish a standardized persistence layer that enables consistent and efficient data management throughout the application.

2.2.2 Notification and Activity Tracking Functions

To improve user awareness and enhance interaction with the system, ProjectPulse incorporates lightweight notification and activity-tracking mechanisms. These functions provide real-time feedback regarding important events and maintain a historical record of user actions.

Notification Function

The `toastNotify(message, type)` function is responsible for generating notification messages within the application.

Each notification is stored as a structured data object containing attributes such as message content, notification type, timestamp, and read/unread status. The stored notifications are subsequently displayed within the Notifications panel, allowing users to review recent system events.

The key objectives of this function are:

- Providing immediate feedback for user actions.
- Recording important system events.
- Maintaining notification history.

Activity Logging Function

The `logActivity(activity)` function records significant operations performed within the application. Examples include project creation, task assignment, status updates, completion events, and workflow modifications. The recorded activities are displayed within the Dashboard module under the “Recent Activities” section, enabling users to monitor project-related events effectively.

The primary functions of the activity logging mechanism include:

- Tracking user and system actions.
- Maintaining an activity history for monitoring purposes.
- Supporting project transparency and traceability.
- Providing real-time updates within the dashboard interface.

By integrating notification management and activity tracking functionalities, ProjectPulse enhances usability and improves situational awareness while maintaining the simplicity of a fully client-side implementation.

2.3 User Interface Implementation

The user interface of ProjectPulse is designed to provide an intuitive, responsive, and visually appealing environment for managing project-related activities. The interface follows modern web design principles and adopts a modular layout structure to ensure consistency, usability, and ease of navigation across all functional components of the application.

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A multi-page architecture is employed to separate different project management functionalities into dedicated modules while maintaining a unified design language throughout the system. This approach enhances user experience by reducing interface complexity and providing clear access to specific project management operations.

2.3.1 Web Interface Implementation

The front-end interface of ProjectPulse is developed using contemporary web technologies that collectively contribute to responsiveness, accessibility, and visual consistency.

HTML5

HTML5 serves as the structural foundation of the application. It defines the layout and organization of all interface elements, including navigation menus, content sections, data tables, forms, dashboard components, and workflow management areas.

CSS3

CSS3 is utilized to implement the visual appearance of the system. Custom styling techniques are applied to create a modern glassmorphism-inspired user interface featuring transparent panels, blur effects, smooth transitions, and aesthetically pleasing visual elements. These design characteristics contribute to an enterprise-grade user experience.

Bootstrap 5

Bootstrap 5 is integrated to provide responsive layouts and standardized user interface components. The framework ensures compatibility across multiple screen sizes and devices while simplifying the implementation of grids, cards, navigation panels, tables, modals, and form controls.

Chart.js

Chart.js is integrated within the Reports Module to generate dynamic and interactive graphical representations of project data. The library supports the visualization of project progress, task distributions, productivity trends, and performance metrics through charts and statistical summaries.

The navigation structure is implemented using a persistent sidebar menu that remains consistent across all application pages. This design enables users to switch seamlessly between modules while maintaining familiarity with the overall system layout. Each page contains dedicated content sections designed specifically for the functionality of the corresponding module.

2.3.2 Internal Screen Modules

ProjectPulse is organized into multiple functional screens, each designed to address a specific aspect of project management. These modules collectively provide a comprehensive project tracking environment.

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Dashboard Module

The Dashboard serves as the primary monitoring interface of the application. It provides users with a consolidated view of key project metrics, including project counts, task statistics, completion rates, and team information. The dashboard also displays recent activities, notifications, and upcoming deadlines, enabling users to quickly assess project status and performance.

Sample Program :

dashboard.html

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <title>Dashboard - Project Management System</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
  <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css"
rel="stylesheet">
  <link rel="stylesheet" href="style.css" />
</head>
<body data-page="dashboard" data-page-attrs>
  <div class="container-app">
    <aside class="sidebar" id="app_sidebar">
      <div class="brand">
        <div class="logo"><i class="fa-solid fa-diagram-project"></i></div>
        <div class="title">
          <strong>ProjectPulse</strong>
          <span>Premium PM SaaS</span>
        </div>
      </div>
      <div class="nav">
        <div class="item active" id="nav_dashboard" data-page="dashboard.html"><div class="ico"><i
class="fa-solid fa-house"></i></div><div class="txt">Dashboard</div></div>
        <div class="item" id="nav_projects" data-page="projects.html"><div class="ico"><i class="fa-
solid fa-folder-open"></i></div><div class="txt">Projects</div></div>
        <div class="item" id="nav_tasks" data-page="tasks.html"><div class="ico"><i class="fa-solid fa-
list-check"></i></div><div class="txt">Tasks</div></div>
```

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```
<div class="item" id="nav_kanban" data-page="kanban.html"><div class="ico"><i class="fa-solid fa-layer-group"></i></div><div class="txt">Kanban Board</div></div>
<div class="item" id="nav_calendar" data-page="calendar.html"><div class="ico"><i class="fa-regular fa-calendar"></i></div><div class="txt">Calendar</div></div>
<div class="item" id="nav_team" data-page="team.html"><div class="ico"><i class="fa-solid fa-users"></i></div><div class="txt">Team Members</div></div>
<div class="item" id="nav_reports" data-page="reports.html"><div class="ico"><i class="fa-solid fa-chart-pie"></i></div><div class="txt">Reports</div></div>
<div class="item" id="nav_settings" data-page="settings.html"><div class="ico"><i class="fa-solid fa-sliders"></i></div><div class="txt">Settings</div></div>
<div class="item" id="nav_help" data-page="about.html"><div class="ico"><i class="fa-solid fa-circle-question"></i></div><div class="txt">Help Center</div></div>
</div>
<div class="bottom">
<div class="theme-toggle">
<div class="label"><strong>Theme</strong><span>Light / Dark</span></div>
<label class="switch mb-0"><input type="checkbox" id="theme_toggle_checkbox" /><span class="slider"></span></label>
</div> </div>
</aside>
<main class="main">
<div class="topbar">
<button class="hamburger" id="sidebar_hamburger" aria-label="Toggle sidebar"><i class="fa-solid fa-bars"></i></button>
<div class="left">
<div style="display:flex; flex-direction:column; line-height:1.1;">
<strong style="font-size:14px; letter-spacing:.2px;">Welcome back</strong>
<span style="font-size:12px; color:var(--muted); font-weight:800;">Manage projects with clarity</span>
</div>
</div>
<div class="search"><i class="fa-solid fa-magnifying-glass"></i><input type="text" placeholder="Search projects, tasks, status... (demo)" /></div>
<div class="right">
<button class="icon-btn" title="Notifications"><i class="fa-solid fa-bell"></i><span class="badge-dot" id="notif_dot" style="display:none;"></span></button>
<button class="btn secondary" id="quick_add_project"><i class="fa-solid fa-plus me-2"></i> Add Project</button>
</div>
</div>
```

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</div>

<section class="section" data-page-content>

<div class="grid dashboard-cards">

<div class="stat-card total" style="grid-column: span 3;"><div class="row"><div class="left"><div class="icon-box"><i class="fa-solid fa-diagram-project"></i></div><div class="meta"><div class="k">Total Projects</div><div class="v" id="stat_total_projects">0</div></div></div><div class="progress"><div class="bar"></div><div class="pct">Live</div></div></div>

<div class="stat-card active" style="grid-column: span 3;"><div class="row"><div class="left"><div class="icon-box"><i class="fa-solid fa-bolt"></i></div><div class="meta"><div class="k">Active Projects</div><div class="v" id="stat_active_projects">0</div></div></div><div class="progress"><div class="bar"> </div><div class="pct">Now</div></div></div>

<div class="stat-card completed" style="grid-column: span 3;"><div class="row"><div class="left"><div class="icon-box"><i class="fa-solid fa-circle-check"></i></div><div class="meta"><div class="k">Completed Projects</div><div class="v" id="stat_completed_projects">0</div></div></div><div class="progress"><div class="bar"> </div><div class="pct">Done</div></div></div>

<div class="stat-card delayed" style="grid-column: span 3;"><div class="row"><div class="left"><div class="icon-box"><i class="fa-solid fa-triangle-exclamation"></i></div><div class="meta"><div class="k">Delayed Projects</div><div class="v" id="stat_delayed_projects">0</div></div></div><div class="progress"><div class="bar"> </div><div class="pct">Risk</div></div></div>

<div class="stat-card pending" style="grid-column: span 3;"><div class="row"><div class="left"><div class="icon-box"><i class="fa-solid fa-clock"></i></div><div class="meta"><div class="k">Total Tasks</div><div class="v" id="stat_total_tasks">0</div></div></div><div class="progress"><div class="bar"> </div><div class="pct">Tasks</div></div></div>

<div class="stat-card completed" style="grid-column: span 3;"><div class="row"><div class="left"><div class="icon-box"><i class="fa-solid fa-check-double"></i></div><div class="meta"><div class="k">Completed Tasks</div><div class="v" id="stat_completed_tasks">0</div></div></div><div class="progress"><div class="bar"> </div><div class="pct">Done</div></div></div>

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```
<div class="stat-card pending" style="grid-column: span 3;"><div class="row"><div class="left">
<div class="icon-box"><i class="fa-solid fa-list"></i></div><div class="meta"><div
class="k">Pending Tasks</div><div class="v" id="stat_pending_tasks">0</div></div></div>
<div class="progress"><div class="bar"><span style="background: linear-gradient(90deg,
rgba(245,158,11,1), rgba(239,68,68,1));" id="stat_pending_tasks_bar">&nbsp;</span></div><div
class="pct">Due</div></div></div>
```

```
<div class="stat-card active" style="grid-column: span 3;"><div class="row"><div class="left">
<div class="icon-box"><i class="fa-solid fa-users"></i></div><div class="meta"><div
class="k">Team Members</div><div class="v" id="stat_team_members">0</div></div></div>
<div class="progress"><div class="bar"><span style="background: linear-gradient(90deg,
rgba(6,182,212,1), rgba(37,99,235,1));" id="stat_team_members_bar">&nbsp;</span></div><div
class="pct">Team</div></div></div>
</div>
```

```
<div class="row g-4 mt-2">
<div class="col-12 col-lg-7">
<div class="panel">
<div class="hd"><h3>Project Completion</h3><span>Progress across tasks</span></div>
<div class="body pb-3">
<div style="display:flex; align-items:center; justify-content:space-between; gap:14px; padding:0 0
8px;">
<div>
<div style="font-size:13px; color:var(--muted); font-weight:900;">Completion Rate</div>
<div style="font-size:28px; font-weight:1000; letter-spacing:-.5px;"
id="project_completion_pct">0%</div>
</div>
<div style="min-width:220px; flex:1;">
<div class="progress"><div class="bar" style="height:10px;"><span id="project_completion_bar"
style="width:0%"></span></div><div class="pct">Updated</div></div>
</div>
</div>
<div style="height:240px; display:flex; align-items:center; justify-content:center; color:var(--
muted); font-weight:900;">
<i class="fa-solid fa-chart-column" style="font-size:34px;"></i>
<div style="margin-left:12px;">Charts render in Reports</div>
</div>
</div>
</div>
</div>
```

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```
<div class="col-12 col-lg-5">
  <div class="panel"><div class="hd"><h3>Recent Activities</h3><span>System timeline</span>
</div><div class="body"><div class="list" id="recent_activities_list"></div></div></div>
  <div class="panel mt-4"><div class="hd"><h3>Upcoming Deadlines</h3><span>Next due
tasks</span></div><div class="body"><div class="list" id="upcoming_deadlines_list"></div>
</div></div>
  <div class="panel mt-4"><div class="hd"><h3>Notifications</h3><span>Read/Unread</span>
</div><div class="body"><div class="list" id="notifications_list"></div></div></div>
</div>
</div> </section> </main> </div>
```

```
<script>
const items = document.querySelectorAll('.sidebar .item[data-page]');
items.forEach(it=>it.addEventListener('click', ()=> location.href = it.getAttribute('data-page')));
const hb = document.getElementById('sidebar_hamburger');
const sb = document.getElementById('app_sidebar');
if(hb && sb) hb.addEventListener('click', ()=> sb.classList.toggle('open'));
</script>
```

```
<script src="script.js"></script>
<footer style="margin: left 40px; padding: 18px 22px 24px;">
  <div class="panel" style="border-radius: 16px; padding: 14px 16px; background:
rgba(255,255,255,.45);">
<div style="display: flex; flex-wrap: wrap; gap: 10px; justify-content: flex-start; align-items: center;
padding-left: 50%; transform: translateX(-50%);">
  <div style="color: var(--muted); font-weight: 900; font-size: 12px;">
Mini Project Developed By:
  <span class="kbd">23AK1A3621 (Janaka Tejeswar)</span>
  <span class="kbd">23AK1A3632 (Rohith)</span>
  <span class="kbd">24AK5A3604 (Nani)</span>
</div>
  <div style="color: var(--muted); font-weight: 900; font-size: 12px;">
Note: Still features are under development.
</div>
</div>
</div>
</div>
</footer>
</body>
</html>
```

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Projects Module

The Projects Module facilitates the management of project records through an organized tabular interface. Users can view and manage project-related information, including project categories, priorities, statuses, schedules, assigned managers, and progress indicators. The module supports efficient project tracking and lifecycle management.

Sample Program :

projects.html

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <title>Projects - Project Management System</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
  <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css"
rel="stylesheet">
  <link rel="stylesheet" href="style.css" />
</head>
<body data-page="projects">
  <div class="container-app">
    <aside class="sidebar" id="app_sidebar">
      <div class="brand">
        <div class="logo"><i class="fa-solid fa-diagram-project"></i></div>
        <div class="title">
          <strong>ProjectPulse</strong>
          <span>Premium PM SaaS</span>
        </div> </div>

      <div class="nav">
        <div class="item" id="nav_dashboard" data-page="dashboard.html"><div class="ico"><i
class="fa-solid fa-house"></i></div><div class="txt">Dashboard</div></div>
        <div class="item active" id="nav_projects" data-page="projects.html"><div class="ico"><i
class="fa-solid fa-folder-open"></i></div><div class="txt">Projects</div></div>
        <div class="item" id="nav_tasks" data-page="tasks.html"><div class="ico"><i class="fa-solid fa-
list-check"></i></div><div class="txt">Tasks</div></div>
```

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```
<div class="item" id="nav_kanban" data-page="kanban.html"><div class="ico"><i class="fa-solid fa-layer-group"></i></div><div class="txt">Kanban Board</div></div>
<div class="item" id="nav_calendar" data-page="calendar.html"><div class="ico"><i class="fa-regular fa-calendar"></i></div><div class="txt">Calendar</div></div>
<div class="item" id="nav_team" data-page="team.html"><div class="ico"><i class="fa-solid fa-users"></i></div><div class="txt">Team Members</div></div>
<div class="item" id="nav_reports" data-page="reports.html"><div class="ico"><i class="fa-solid fa-chart-pie"></i></div><div class="txt">Reports</div></div>
<div class="item" id="nav_settings" data-page="settings.html"><div class="ico"><i class="fa-solid fa-sliders"></i></div><div class="txt">Settings</div></div>
<div class="item" id="nav_help" data-page="about.html"><div class="ico"><i class="fa-solid fa-circle-question"></i></div><div class="txt">Help Center</div></div>
</div>
```

```
<div class="bottom">
<div class="theme-toggle">
<div class="label"><strong>Theme</strong><span>Light / Dark</span></div>
<label class="switch mb-0">
<input type="checkbox" id="theme_toggle_checkbox" />
<span class="slider"></span>
</label>
</div>
</div>
</aside>
```

```
<main class="main">
<div class="topbar">
<button class="hamburger" id="sidebar_hamburger" aria-label="Toggle sidebar"><i class="fa-solid fa-bars"></i></button>
```

```
<div class="left">
<div style="display:flex; flex-direction:column; line-height:1.1;">
<strong style="font-size:14px; letter-spacing:.2px;">Projects</strong>
<span style="font-size:12px; color:var(--muted); font-weight:800;">Manage planning, delivery, and progress</span>
</div>
</div>
```

```
<div class="search">
```

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```
<i class="fa-solid fa-magnifying-glass"></i>
```

```
<input type="text" placeholder="Search projects... (demo)" />
```

```
</div>
```

```
<div class="right">
```

```
<button class="icon-btn" title="Notifications"><i class="fa-solid fa-bell"></i><span class="badge-dot" id="notif_dot" style="display:none;"></span></button>
```

```
<button class="btn secondary" disabled><i class="fa-solid fa-plus me-2"></i> Add Project</button>
```

```
</div>
```

```
</div>
```

```
<section class="section">
```

```
<div class="panel">
```

```
<div class="hd">
```

```
<h3>Projects Table</h3>
```

```
<span>LocalStorage data • Actions UI (demo)</span>
```

```
</div>
```

```
<div class="body">
```

```
<div class="table-wrap">
```

```
<table class="table mb-0">
```

```
<thead>
```

```
<tr>
```

```
<th style="width:120px;">Project ID</th>
```

```
<th>Project</th>
```

```
<th style="width:160px;">Category</th>
```

```
<th style="width:130px;">Priority</th>
```

```
<th style="width:140px;">Status</th>
```

```
<th style="width:130px;">Start</th>
```

```
<th style="width:130px;">End</th>
```

```
<th style="width:120px;">Progress</th>
```

```
<th style="width:220px;">Actions</th>
```

```
</tr>
```

```
</thead>
```

```
<tbody id="projects_table_body">
```

```
<tr><td colspan="9" style="color:var(--muted); font-weight:900;">Loading projects...</td></tr>
```

```
</tbody>
```

```
</table>
```

```
</div>
```

Project Management System

```
<div class="mt-3 d-flex flex-wrap gap-2" style="justify-content:space-between; align-items:center;">
<div style="color:var(--muted); font-weight:900; font-size:12px;">
  Tip: Create/manage tasks from <span class="kbd"><i class="fa-solid fa-list-check"></i>
Tasks</span> & visualize via <span class="kbd"><i class="fa-solid fa-layer-group"></i>
Kanban</span>.
</div>
<div class="d-flex gap-2">
  <a class="btn secondary" href="kanban.html"><i class="fa-solid fa-layer-group me-2"></i>
Kanban</a>
  <a class="btn secondary" href="reports.html"><i class="fa-solid fa-chart-simple me-2"></i>
Reports</a>
</div>
</div>
</div>
</div>
</div>
</section>
</main>
</div>

<script>
const items = document.querySelectorAll('.sidebar .item[data-page]');
items.forEach(it => it.addEventListener('click', () => location.href = it.getAttribute('data-page')));
const hb = document.getElementById('sidebar_hamburger');
const sb = document.getElementById('app_sidebar');
if(hb && sb) hb.addEventListener('click', () => sb.classList.toggle('open'));

window.addEventListener('DOMContentLoaded', () => {
if (typeof renderProjectsTable === 'function') renderProjectsTable();
});
</script>

<script src="script.js"></script>

<footer style="margin-left:0; padding:18px 22px 24px;">
  <div class="panel" style="border-radius:16px; padding:14px 16px; background:
rgba(255,255,255,.45);">
<div style="display:flex; flex-wrap:wrap; gap:10px; justify-content:flex-start; align-items:center;
padding-left:50%; transform:translateX(-50%);">
```

Project Management System

<div style="color:var(--muted); font-weight:900; font-size:12px;">

Mini Project Developed By:

23AK1A3621 (Janaka Tejeswar)

23AK1A3632 (Rohith)

24AK5A3604 (Nani)

</div>

<div style="color:var(--muted); font-weight:900; font-size:12px;">

Note: Still features are under development.

</div>

</div>

</div>

</footer>

</body>

</html>

Project Management System

Tasks Module

The Tasks Module provides a structured environment for task management. Tasks are displayed within interactive tables containing details such as deadlines, priority levels, assignment information, completion status, categories, and progress percentages. This module assists users in monitoring work assignments and project execution activities.

Sample Program :

tasks.html

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <title>Tasks - Project Management System</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
  <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css"
rel="stylesheet">
  <link rel="stylesheet" href="style.css" />
</head>
<body data-page="tasks">
  <div class="container-app">
    <aside class="sidebar" id="app_sidebar">
      <div class="brand">
        <div class="logo"><i class="fa-solid fa-diagram-project"></i></div>
        <div class="title">
          <strong>ProjectPulse</strong>
          <span>Premium PM SaaS</span>
        </div>
      </div>
      <div class="nav">
        <div class="item" id="nav_dashboard" data-page="dashboard.html"><div class="ico"><i
class="fa-solid fa-house"></i></div><div class="txt">Dashboard</div></div>
        <div class="item" id="nav_projects" data-page="projects.html"><div class="ico"><i class="fa-
solid fa-folder-open"></i></div><div class="txt">Projects</div></div>
        <div class="item active" id="nav_tasks" data-page="tasks.html"><div class="ico"><i class="fa-
solid fa-list-check"></i></div><div class="txt">Tasks</div></div>
```

Project Management System

```
<div class="item" id="nav_kanban" data-page="kanban.html"><div class="ico"><i class="fa-solid fa-layer-group"></i></div><div class="txt">Kanban Board</div></div>
<div class="item" id="nav_calendar" data-page="calendar.html"><div class="ico"><i class="fa-regular fa-calendar"></i></div><div class="txt">Calendar</div></div>
<div class="item" id="nav_team" data-page="team.html"><div class="ico"><i class="fa-solid fa-users"></i></div><div class="txt">Team Members</div></div>
<div class="item" id="nav_reports" data-page="reports.html"><div class="ico"><i class="fa-solid fa-chart-pie"></i></div><div class="txt">Reports</div></div>
<div class="item" id="nav_settings" data-page="settings.html"><div class="ico"><i class="fa-solid fa-sliders"></i></div><div class="txt">Settings</div></div>
<div class="item" id="nav_help" data-page="about.html"><div class="ico"><i class="fa-solid fa-circle-question"></i></div><div class="txt">Help Center</div></div>
</div>
```

```
<div class="bottom">
<div class="theme-toggle">
<div class="label"><strong>Theme</strong><span>Light / Dark</span></div>
<label class="switch mb-0">
<input type="checkbox" id="theme_toggle_checkbox" />
<span class="slider"></span>
</label>
</div>
</div>
</aside>
```

```
<main class="main">
<div class="topbar">
<button class="hamburger" id="sidebar_hamburger" aria-label="Toggle sidebar"><i class="fa-solid fa-bars"></i></button>
```

```
<div class="left">
<div style="display:flex; flex-direction:column; line-height:1.1;">
<strong style="font-size:14px; letter-spacing:.2px;">Tasks</strong>
<span style="font-size:12px; color:var(--muted); font-weight:800;">Track work items & progress</span>
</div>
</div>
```

```
<div class="search">
```

Project Management System

```
<i class="fa-solid fa-magnifying-glass"></i>
<input type="text" placeholder="Search tasks... (demo)" />
</div>

<div class="right">
<button class="icon-btn" title="Notifications">
<i class="fa-solid fa-bell"></i>
<span class="badge-dot" id="notif_dot" style="display:none;"></span>
</button>
<button class="btn secondary" disabled><i class="fa-solid fa-plus me-2"></i> Add Task</button>
</div>
</div>

<section class="section">
<div class="panel">
<div class="hd">
<h3>Task List</h3>
<span>Drag & drop status in Kanban</span>
</div>

<div class="body">
<div class="table-wrap">
<table class="table mb-0">
<thead>
<tr>
<th style="width:110px;">Task ID</th>
<th>Task</th>
<th style="width:120px;">Priority</th>
<th style="width:120px;">Deadline</th>
<th style="width:140px;">Status</th>
<th style="width:160px;">Assigned</th>
<th style="width:150px;">Category</th>
<th style="width:110px;">Progress</th>
<th style="width:220px;">Actions</th>
</tr>
</thead>
<tbody id="tasks_table_body">
<tr><td colspan="9" style="color:var(--muted); font-weight:900;">Loading tasks...</td></tr>
</tbody>
```

Project Management System

</table>

</div>

<div class="mt-3 d-flex flex-wrap gap-2" style="align-items:center; justify-content:space-between;">

<div style="color:var(--muted); font-weight:900; font-size:12px;">

Tip: Update status using <i class="fa-solid fa-layer-group"></i> Kanban

</div>

<div class="d-flex gap-2">

<i class="fa-solid fa-route me-2"></i> Go to Kanban

</div>

</div>

</div>

</div>

</section>

</main>

</div>

<script>

const items = document.querySelectorAll('.sidebar .item[data-page]');

items.forEach(it => it.addEventListener('click', () => location.href = it.getAttribute('data-page')));

const hb = document.getElementById('sidebar_hamburger');

const sb = document.getElementById('app_sidebar');

if (hb && sb) hb.addEventListener('click', () => sb.classList.toggle('open'));

window.addEventListener('DOMContentLoaded', () => {

if (typeof renderTasksTable === 'function') renderTasksTable();

});

</script>

<script src="script.js"></script>

<footer style="margin-left:0; padding:18px 22px 24px;">

<div class="panel" style="border-radius:16px; padding:14px 16px; background:rgba(255,255,255,.45);">

<div style="display:flex; flex-wrap:wrap; gap:10px; justify-content:flex-start; align-items:center; padding-left:50%; transform:translateX(-50%);">

<div style="color:var(--muted); font-weight:900; font-size:12px;">

Project Management System

Mini Project Developed By:

23AK1A3621 (Janaka Tejeswar)

23AK1A3632 (Rohith)

24AK5A3604 (Nani)

</div>

<div style="color:var(--muted); font-weight:900; font-size:12px;">

Note: Still features are under development.

</div>

</div>

</div>

</footer>

</body>

</html>

Kanban Board Module

The Kanban Board offers a visual workflow management interface based on task lifecycle stages. Tasks are represented as movable cards positioned within workflow columns. Users can perform drag-and-drop operations to update task statuses, enabling efficient workflow monitoring and task progression management.

Sample Program :

tasks.html

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <title>Kanban Board - Project Management System</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
  <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css"
rel="stylesheet">
  <link rel="stylesheet" href="style.css" />
</head>
<body data-page="kanban">
  <div class="container-app">
    <aside class="sidebar" id="app_sidebar">
      <div class="brand">
        <div class="logo"><i class="fa-solid fa-diagram-project"></i></div>
        <div class="title"><strong>ProjectPulse</strong><span>Premium PM SaaS</span></div>
      </div>

      <div class="nav">
        <div class="item" id="nav_dashboard" data-page="dashboard.html"><div class="ico"><i
class="fa-solid fa-house"></i></div><div class="txt">Dashboard</div></div>
        <div class="item" id="nav_projects" data-page="projects.html"><div class="ico"><i class="fa-
solid fa-folder-open"></i></div><div class="txt">Projects</div></div>
        <div class="item" id="nav_tasks" data-page="tasks.html"><div class="ico"><i class="fa-solid fa-
list-check"></i></div><div class="txt">Tasks</div></div>
        <div class="item active" id="nav_kanban" data-page="kanban.html"><div class="ico"><i
class="fa-solid fa-layer-group"></i></div><div class="txt">Kanban Board</div></div>
```

Project Management System

```
<div class="item" id="nav_calendar" data-page="calendar.html"><div class="ico"><i class="fa-regular fa-calendar"></i></div><div class="txt">Calendar</div></div>
<div class="item" id="nav_team" data-page="team.html"><div class="ico"><i class="fa-solid fa-users"></i></div><div class="txt">Team Members</div></div>
<div class="item" id="nav_reports" data-page="reports.html"><div class="ico"><i class="fa-solid fa-chart-pie"></i></div><div class="txt">Reports</div></div>
<div class="item" id="nav_settings" data-page="settings.html"><div class="ico"><i class="fa-solid fa-sliders"></i></div><div class="txt">Settings</div></div>
<div class="item" id="nav_help" data-page="about.html"><div class="ico"><i class="fa-solid fa-circle-question"></i></div><div class="txt">Help Center</div></div>
</div>
```

```
<div class="bottom">
<div class="theme-toggle">
<div class="label"><strong>Theme</strong><span>Light / Dark</span></div>
  <label class="switch mb-0"><input type="checkbox" id="theme_toggle_checkbox" /><span
class="slider"></span></label>
</div>
</div>
</aside>
```

```
<main class="main">
<div class="topbar">
  <button class="hamburger" id="sidebar_hamburger" aria-label="Toggle sidebar"><i class="fa-solid fa-bars"></i></button>
  <div class="left">
    <div style="display:flex; flex-direction:column; line-height:1.1;">
      <strong style="font-size:14px; letter-spacing:.2px;">Kanban Board</strong>
      <span style="font-size:12px; color:var(--muted); font-weight:800;">Drag tasks across columns</span>
    </div>
  </div>
  <div class="search"><i class="fa-solid fa-magnifying-glass"></i><input type="text"
placeholder="Search tasks... (demo)" /></div>
  <div class="right">
    <button class="icon-btn" title="Notifications"><i class="fa-solid fa-bell"></i><span class="badge-dot" id="notif_dot" style="display:none;"></span></button>
    <a class="btn primary" href="tasks.html"><i class="fa-solid fa-plus me-2"></i> Add Task</a>
  </div>
```

Project Management System

</div>

<section class="section">

<div class="panel mb-4">

<div class="hd">

<h3>Task Columns</h3>

Real-time UI updates (LocalStorage)

</div>

<div class="body">

<div class="list-item" style="margin:0;">

<div class="li-left">

<div class="ico"><i class="fa-solid fa-hand-pointer"></i></div>

<div class="txt">

Tip

<small>Drag any task card to update its status. Completed tasks auto-set progress to 100%.</small>

</div>

</div>

<div class="li-right">Board: To Do → In Progress → Review → Testing → Completed</div>

</div>

</div>

</div>

<div class="kanban">

<div class="col todo" id="kanban_todo" aria-label="To Do column">

<div class="thead"> TO DO<span

class="kbd">Drag</div>

<div class="tbody"></div>

</div>

<div class="col inprog" id="kanban_inprogress" aria-label="In Progress column">

<div class="thead"> IN PROGRESS<span

class="kbd">Work</div>

<div class="tbody"></div>

</div>

<div class="col review" id="kanban_review" aria-label="Review column">

<div class="thead"> REVIEW<span

class="kbd">Check</div>

<div class="tbody"></div>

</div>

<div class="col testing" id="kanban_testing" aria-label="Testing column">

Project Management System

```
<div class="thead"><strong><span class="dot"></span>TESTING</strong><span
class="kbd">Verify</span></div>
<div class="tbody"></div>
</div>
<div class="col done" id="kanban_done" aria-label="Completed column">
  <div class="thead"><strong><span class="dot"></span>COMPLETED</strong><span
class="kbd">Done</span></div>
  <div class="tbody"></div>
</div> </div> </section> </main> </div>

<script>
const items = document.querySelectorAll('.sidebar .item[data-page]');
items.forEach(it => it.addEventListener('click', () => location.href = it.getAttribute('data-page')));
const hb = document.getElementById('sidebar_hamburger');
const sb = document.getElementById('app_sidebar');
if(hb && sb) hb.addEventListener('click', () => sb.classList.toggle('open'));

window.addEventListener('DOMContentLoaded', () => {
if(typeof renderKanban === 'function') renderKanban();
});
</script> <script src="script.js"></script>

<footer style="margin-left:0; padding:18px 22px 24px;">
  <div class="panel" style="border-radius:16px; padding:14px 16px; background:
rgba(255,255,255,.45);">
<div style="display:flex; flex-wrap:wrap; gap:10px; justify-content:flex-start; align-items:center;
padding-left:50%; transform:translateX(-50%);">
  <div style="color:var(--muted); font-weight:900; font-size:12px;">
Mini Project Developed By:
  <span class="kbd">23AK1A3621 (Janaka Tejeswar)</span>
  <span class="kbd">23AK1A3632 (Rohith)</span>
  <span class="kbd">24AK5A3604 (Nani)</span>
  </div>
  <div style="color:var(--muted); font-weight:900; font-size:12px;">
Note: Still features are under development.
  </div> </div> </div>
</footer>
</body>
</html>
```

Calendar Module

The Calendar Module presents project schedules and task deadlines within a monthly calendar view. Important dates and upcoming deadlines are highlighted to improve schedule visibility and facilitate proactive project planning. The calendar serves as a deadline-awareness tool that assists users in managing time-sensitive activities.

Sample Program :

tasks.html

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <title>Calendar - Project Management System</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
  <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css"
rel="stylesheet">
  <link rel="stylesheet" href="style.css" />
</head>
<body data-page="calendar">
  <div class="container-app">
    <aside class="sidebar" id="app_sidebar">
      <div class="brand">
        <div class="logo"><i class="fa-solid fa-diagram-project"></i></div>
        <div class="title"><strong>ProjectPulse</strong><span>Premium PM SaaS</span></div>
      </div>

      <div class="nav">
        <div class="item" id="nav_dashboard" data-page="dashboard.html"><div class="ico"><i
class="fa-solid fa-house"></i></div><div class="txt">Dashboard</div></div>
        <div class="item" id="nav_projects" data-page="projects.html"><div class="ico"><i class="fa-
solid fa-folder-open"></i></div><div class="txt">Projects</div></div>
        <div class="item" id="nav_tasks" data-page="tasks.html"><div class="ico"><i class="fa-solid fa-
list-check"></i></div><div class="txt">Tasks</div></div>
        <div class="item" id="nav_kanban" data-page="kanban.html"><div class="ico"><i class="fa-solid
fa-layer-group"></i></div><div class="txt">Kanban Board</div></div>
      </div>
    </aside>
  </div>
</body>
</html>
```


Project Management System

```
<div class="item active" id="nav_calendar" data-page="calendar.html"><div class="ico"><i
class="fa-regular fa-calendar"></i></div><div class="txt">Calendar</div></div>
<div class="item" id="nav_team" data-page="team.html"><div class="ico"><i class="fa-solid fa-
users"></i></div><div class="txt">Team Members</div></div>
<div class="item" id="nav_reports" data-page="reports.html"><div class="ico"><i class="fa-solid
fa-chart-pie"></i></div><div class="txt">Reports</div></div>
<div class="item" id="nav_settings" data-page="settings.html"><div class="ico"><i class="fa-solid
fa-sliders"></i></div><div class="txt">Settings</div></div>
<div class="item" id="nav_help" data-page="about.html"><div class="ico"><i class="fa-solid fa-
circle-question"></i></div><div class="txt">Help Center</div></div>
</div>
```

```
<div class="bottom">
<div class="theme-toggle">
<div class="label"><strong>Theme</strong><span>Light / Dark</span></div>
<label class="switch mb-0"><input type="checkbox" id="theme_toggle_checkbox" /><span
class="slider"></span></label>
</div>
</div>
</aside>
```

```
<main class="main">
<div class="topbar">
<button class="hamburger" id="sidebar_hamburger" aria-label="Toggle sidebar"><i class="fa-solid
fa-bars"></i></button>
```

```
<div class="left">
<div style="display:flex; flex-direction:column; line-height:1.1;">
<strong style="font-size:14px; letter-spacing:.2px;">Calendar</strong>
<span style="font-size:12px; color:var(--muted); font-weight:800;">Month view • deadlines</span>
</div>
</div>
```

```
<div class="search">
<i class="fa-solid fa-magnifying-glass"></i>
<input type="text" placeholder="Search events... (demo)" />
</div>
```

```
<div class="right">
```

Project Management System

```
<button class="icon-btn" title="Notifications"><i class="fa-solid fa-bell"></i><span class="badge-dot" id="notif_dot" style="display:none;"></span></button>
```

```
<button class="btn secondary" disabled><i class="fa-solid fa-plus me-2"></i> Add Event</button>
```

```
</div>
```

```
</div>
```

```
<section class="section">
```

```
<div class="row g-4">
```

```
<div class="col-12 col-lg-8">
```

```
<div class="panel">
```

```
<div class="hd"><h3>Project Calendar</h3><span id="calendar_range_label">—</span></div>
```

```
<div class="body">
```

```
<div class="d-flex align-items-center justify-content-between flex-wrap gap-2 mb-3">
```

```
<div class="d-flex gap-2">
```

```
<button class="sm-btn" id="cal_prev"><i class="fa-solid fa-chevron-left me-2"></i>Prev</button>
```

```
<button class="sm-btn" id="cal_today"><i class="fa-solid fa-bullseye me-2"></i>Today</button>
```

```
<button class="sm-btn" id="cal_next">Next<i class="fa-solid fa-chevron-right ms-2"></i></button>
```

```
</div>
```

```
<div class="d-flex gap-2">
```

```
<span class="kbd"><i class="fa-solid fa-calendar-days"></i> Month</span>
```

```
<span class="kbd"><i class="fa-solid fa-clock"></i> Deadline highlights</span>
```

```
</div>
```

```
</div>
```

```
<div class="table-wrap">
```

```
<table class="table mb-0" style="min-width:720px;">
```

```
<thead>
```

```
<tr><th>Sun</th><th>Mon</th><th>Tue</th><th>Wed</th><th>Thu</th><th>Fri</th>
```

```
<th>Sat</th></tr>
```

```
</thead>
```

```
<tbody id="calendar_grid"></tbody>
```

```
</table>
```

```
</div>
```

```
<div class="mt-3" style="color:var(--muted); font-weight:900; font-size:12px;">
```

```
Legend: <span class="pill primary" style="padding:5px 10px;">Deadline</span>
```

```
<span class="pill amber" style="padding:5px 10px;">Overdue</span>
```

```
</div>
```

```
</div>
```

Project Management System

</div>

</div>

<div class="col-12 col-lg-4">

<div class="panel">

<div class="hd"><h3>Deadline Reminder</h3>Upcoming</div>

<div class="body"><div class="list" id="calendar_upcoming"></div></div>

</div>

<div class="panel mt-4">

<div class="hd"><h3>Overdue Tasks</h3>Needs attention</div>

<div class="body"><div class="list" id="calendar_overdue"></div></div>

</div>

<div class="panel mt-4">

<div class="hd"><h3>Milestones</h3>Demo events</div>

<div class="body"><div class="list" id="calendar_milestones"></div></div>

</div>

</div>

</div>

</section>

</main>

</div>

<script>

const items = document.querySelectorAll('.sidebar .item[data-page]');

items.forEach(it => it.addEventListener('click', () => location.href = it.getAttribute('data-page')));

const hb = document.getElementById('sidebar_hamburger');

const sb = document.getElementById('app_sidebar');

if(hb && sb) hb.addEventListener('click', () => sb.classList.toggle('open'));

function loadTasks(){

try{ return JSON.parse(localStorage.getItem('pms_tasks')||'[]'); } catch{ return []; }

}

const tasks = loadTasks();

const grid = document.getElementById('calendar_grid');

const label = document.getElementById('calendar_range_label');

const upcomingEl = document.getElementById('calendar_upcoming');

Project Management System

```
const overdueEl = document.getElementById('calendar_overdue');
const milestonesEl = document.getElementById('calendar_milestones');

const deadlineMap = new Map();
tasks.forEach(t => {
  if(!t.deadline) return;
  const key = t.deadline;
  if(!deadlineMap.has(key)) deadlineMap.set(key, []);
  deadlineMap.get(key).push(t);
});

const today = new Date();
let view = new Date(today.getFullYear(), today.getMonth(), 1);

const fmtDay = (d)=> d.toISOString().slice(0,10);

function renderMonth(){
  const y = view.getFullYear();
  const m = view.getMonth();
  const first = new Date(y,m,1);
  const firstDay = first.getDay();
  const weeks = [];

  const startDate = new Date(y,m,1-firstDay);
  for(let w=0; w<6; w++){
    const week = [];
    for(let d=0; d<7; d++){
      const cur = new Date(startDate);
      cur.setDate(startDate.getDate()+w*7+d);
      week.push(cur);
    }
    weeks.push(week);
  }

  if(label) label.textContent = view.toLocaleString([], {month:'long', year:'numeric'});
  if(!grid) return;

  const html = weeks.map(week=>{
    const tds = week.map(day=>{
```

Project Management System

```
const key = fmtDay(day);
const inMonth = day.getMonth()===m;
const list = deadlineMap.get(key) || [];
const overdue = list.some(t => t.deadline && new Date(t.deadline).getTime() < Date.now() && !
['Completed','Cancelled'].includes(t.status));

const cls = overdue ? 'pill amber' : 'pill primary';
const badge = list.length ? `<div style="margin-top:4px; display:flex; justify-content:center;"><span
class="${cls}" style="padding:4px 8px; font-size:11px; font-weight:1000;">${list.length}</span>
</div>` : "";

return `<td style="height:72px; vertical-align:top; opacity:${inMonth?1:.35};">
<div style="font-weight:1000; font-size:12px; color:var(--muted);">${day.getDate()}</div>
${badge}
</td>`;
}).join("");
return `<tr>${tds}</tr>`;
}).join("");

grid.innerHTML = html;
}

function renderUpcoming(){
if(!upcomingEl) return;
const now = Date.now();
const upcoming = tasks
.filter(t=>t.deadline)
.map(t=>({t, dt: new Date(t.deadline)}))
.filter(x => x.dt.getTime() >= now && x.dt.getTime() <= now + 7*24*3600*1000 && !
['Completed','Cancelled'].includes(x.t.status))
.sort((a,b)=>a.dt-b.dt)
.slice(0,8)
.map(x=>x.t);

upcomingEl.innerHTML = upcoming.length
? upcoming.map(t=>`
<div class="list-item">
<div class="li-left">
<div class="ico"><i class="fa-regular fa-clock"></i></div>
```

Project Management System

```
<div class="txt">
<strong>${t.name}</strong>
<small>Deadline: ${t.deadline}</small>
<small>${t.status}</small>
</div>
</div>
<div class="li-right"><span class="pill amber" style="padding:5px 10px;">Soon</span></div>
</div>
`).join("")
: '<div class="list-item"><div class="li-left"><div class="txt"><strong>No upcoming
deadlines</strong><small>You're on track.</small></div></div></div>';
}

function renderOverdue(){
if(!overdueEl) return;
const overdue = tasks
.filter(t=>t.deadline)
.filter(t => ![ 'Completed','Cancelled'].includes(t.status) && new Date(t.deadline).getTime() <
Date.now())
.slice(0,8);
overdueEl.innerHTML = overdue.length
? overdue.map(t=>`
<div class="list-item">
<div class="li-left">
<div class="ico"><i class="fa-solid fa-triangle-exclamation"></i></div>
<div class="txt">
<strong>${t.name}</strong>
<small>Deadline: ${t.deadline}</small>
<small>Status: ${t.status}</small>
</div>
</div>
<div class="li-right"><span class="pill red" style="padding:5px 10px;">Overdue</span></div>
</div>
`).join("")
: '<div class="list-item"><div class="li-left"><div class="txt"><strong>No overdue tasks</strong>
<small>Great work!</small></div></div></div>';
}

function renderMilestones(){
```

Project Management System

```
if(!milestonesEl) return;
const demo = [
  {title:'Sprint Planning', date: new Date(today.getFullYear(), today.getMonth(),
today.getDate()+3).toISOString().slice(0,10)},
  {title:'Design Review', date: new Date(today.getFullYear(), today.getMonth(),
today.getDate()+10).toISOString().slice(0,10)},
  {title:'Release Candidate', date: new Date(today.getFullYear(), today.getMonth()+1,
5).toISOString().slice(0,10)}
];
milestonesEl.innerHTML = demo.map(m=>`
<div class="list-item">
<div class="li-left">
<div class="ico"><i class="fa-solid fa-flag"></i></div>
<div class="txt"><strong>${m.title}</strong><small>${m.date}</small></div>
</div>
<div class="li-right"><span class="pill cyan" style="padding:5px 10px;">Milestone</span></div>
</div>
`).join("");
}
```

```
document.getElementById('cal_prev')?.addEventListener('click', ()=>{
view.setMonth(view.getMonth()-1); renderMonth();});
document.getElementById('cal_next')?.addEventListener('click', ()=>{
view.setMonth(view.getMonth()+1); renderMonth();});
document.getElementById('cal_today')?.addEventListener('click', ()=>{ view = new
Date(today.getFullYear(), today.getMonth(), 1); renderMonth();});
```

```
renderMonth();
renderUpcoming();
renderOverdue();
renderMilestones();
</script>
```

```
<script src="script.js"></script>
<footer style="margin-left:0; padding:18px 22px 24px;">
  <div class="panel" style="border-radius:16px; padding:14px 16px; background:
rgba(255,255,255,.45);">
<div style="display:flex; flex-wrap:wrap; gap:10px; justify-content:flex-start; align-items:center;
padding-left:50%; transform:translateX(-50%);">
```

Project Management System

<div style="color:var(--muted); font-weight:900; font-size:12px;">

Mini Project Developed By:

23AK1A3621 (Janaka Tejeswar)

23AK1A3632 (Rohith)

24AK5A3604 (Nani)

</div>

<div style="color:var(--muted); font-weight:900; font-size:12px;">

Note: Still features are under development.

</div>

</div>

</div>

</footer>

</body>

</html>

Reports Module

The Reports Module provides analytical insights through graphical visualizations and summary statistics. Interactive charts generated using Chart.js enable users to evaluate project performance, monitor completion trends, analyze task distributions, and identify productivity patterns. These analytical capabilities support informed decision-making and project evaluation.

Sample Program :

reports.html

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <title>Reports - Project Management System</title>
  <link      href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
  <link      href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css"
rel="stylesheet">
  <link rel="stylesheet" href="style.css" />
</head>
<body data-page="reports">
  <div class="container-app">
    <aside class="sidebar" id="app_sidebar">
      <div class="brand">
        <div class="logo"><i class="fa-solid fa-diagram-project"></i></div>
        <div class="title"><strong>ProjectPulse</strong><span>Premium PM SaaS</span></div>
      </div>

      <div class="nav">
        <div class="item" id="nav_dashboard" data-page="dashboard.html"><div class="ico"><i
class="fa-solid fa-house"></i></div><div class="txt">Dashboard</div></div>
        <div class="item" id="nav_projects" data-page="projects.html"><div class="ico"><i class="fa-
solid fa-folder-open"></i></div><div class="txt">Projects</div></div>
        <div class="item" id="nav_tasks" data-page="tasks.html"><div class="ico"><i class="fa-solid fa-
list-check"></i></div><div class="txt">Tasks</div></div>
        <div class="item" id="nav_kanban" data-page="kanban.html"><div class="ico"><i class="fa-solid
fa-layer-group"></i></div><div class="txt">Kanban Board</div></div>
      </div>
    </aside>
  </div>
</body>
</html>
```

Project Management System

```
<div class="item" id="nav_calendar" data-page="calendar.html"><div class="ico"><i class="fa-regular fa-calendar"></i></div><div class="txt">Calendar</div></div>
<div class="item" id="nav_team" data-page="team.html"><div class="ico"><i class="fa-solid fa-users"></i></div><div class="txt">Team Members</div></div>
<div class="item active" id="nav_reports" data-page="reports.html"><div class="ico"><i class="fa-solid fa-chart-pie"></i></div><div class="txt">Reports</div></div>
<div class="item" id="nav_settings" data-page="settings.html"><div class="ico"><i class="fa-solid fa-sliders"></i></div><div class="txt">Settings</div></div>
<div class="item" id="nav_help" data-page="about.html"><div class="ico"><i class="fa-solid fa-circle-question"></i></div><div class="txt">Help Center</div></div>
</div>
```

```
<div class="bottom">
<div class="theme-toggle">
<div class="label"><strong>Theme</strong><span>Light / Dark</span></div>
  <label class="switch mb-0"><input type="checkbox" id="theme_toggle_checkbox" /><span
class="slider"></span></label>
</div>
</div>
</aside>
```

```
<main class="main">
<div class="topbar">
  <button class="hamburger" id="sidebar_hamburger" aria-label="Toggle sidebar"><i class="fa-solid fa-bars"></i></button>
  <div class="left">
    <div style="display:flex; flex-direction:column; line-height:1.1;">
      <strong style="font-size:14px; letter-spacing:.2px;">Reports</strong>
      <span style="font-size:12px; color:var(--muted); font-weight:800;">Charts, statistics, exports</span>
    </div>
  </div>
  <div class="search"><i class="fa-solid fa-magnifying-glass"></i><input type="text"
placeholder="Search reports... (demo)" /></div>
  <div class="right">
    <button class="icon-btn" title="Notifications"><i class="fa-solid fa-bell"></i><span class="badge-dot" id="notif_dot" style="display:none;"></span></button>
    <button class="btn primary" id="btn_export_csv"><i class="fa-solid fa-file-csv me-2"></i>Export CSV</button>
```

Project Management System

</div>

</div>

<section class="section">

<div class="row g-4">

<div class="col-12 col-lg-7">

<div class="panel">

<div class="hd">

<h3>Analytics Dashboard</h3>

Chart.js powered

</div>

<div class="body">

<div class="row g-4">

<div class="col-12">

<div style="background:rgba(255,255,255,.35); border:1px solid rgba(226,232,240,.55); border-radius:16px; padding:14px;">

<div style="display:flex; justify-content:space-between; align-items:center; gap:12px; margin-bottom:10px;">

<div>

<div style="font-weight:1000; font-size:14px;">Project Completion</div>

<div style="color:var(--muted); font-weight:900; font-size:12px;">Completed vs Total tasks</div>

</div>

Live

</div>

<canvas id="chart_project_completion" height="110"></canvas>

</div>

</div>

<div class="col-12 col-md-6">

<div style="background:rgba(255,255,255,.35); border:1px solid rgba(226,232,240,.55); border-radius:16px; padding:14px;">

<div style="font-weight:1000; font-size:14px;">Task Status</div>

<div style="color:var(--muted); font-weight:900; font-size:12px;">To Do / In Progress / Review / Testing / Completed</div>

<canvas id="chart_task_status" height="120"></canvas>

</div>

</div>

<div class="col-12 col-md-6">

Project Management System

```
<div style="background:rgba(255,255,255,.35); border:1px solid rgba(226,232,240,.55); border-
radius:16px; padding:14px;">
<div style="font-weight:1000; font-size:14px;">Priority Distribution</div>
<div style="color:var(--muted); font-weight:900; font-size:12px;">Low / Medium / High /
Critical</div>
<canvas id="chart_priority" height="120"></canvas>
</div>
</div>
```

```
<div class="col-12">
<div style="background:rgba(255,255,255,.35); border:1px solid rgba(226,232,240,.55); border-
radius:16px; padding:14px;">
<div style="display:flex; justify-content:space-between; align-items:center; gap:12px; margin-
bottom:10px;">
<div>
<div style="font-weight:1000; font-size:14px;">Monthly Progress (Demo)</div>
<div style="color:var(--muted); font-weight:900; font-size:12px;">Line chart with smooth
tooltips</div>
</div>
<span class="pill cyan">Preview</span>
</div>
<canvas id="chart_monthly" height="120"></canvas>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
```

```
<div class="col-12 col-lg-5">
<div class="panel">
<div class="hd">
<h3>Reports Center</h3>
<span>Summary tables</span>
</div>
<div class="body">
<div class="list" id="reports_summary_list"></div>

<div class="mt-4">
```

Project Management System

```
<div class="hd" style="padding:0 0 12px; border:0;">
<h3 style="font-size:14px; margin:0;">Completion Report</h3>
<span style="color:var(--muted); font-weight:900; font-size:12px;">Success rate</span>
</div>
    <div id="completion_report_box" style="border-radius:16px; border:1px solid
    rgba(226,232,240,.55); background:rgba(255,255,255,.35); padding:14px;">
    <div style="display:flex; justify-content:space-between; gap:12px; align-items:center;">
    <div>
    <div style="font-size:12px; color:var(--muted); font-weight:1000;">Success Rate</div>
    <div style="font-size:30px; font-weight:1000; letter-spacing:-.5px;" id="success_rate_value">0%
    </div>
    </div>
    <div style="min-width:220px; flex:1;">
    <div class="progress"><div class="bar" style="height:10px;"><span id="success_rate_bar"
    style="width:0%"></span></div><div class="pct" id="success_rate_label">—</div></div>
    </div>
    </div>
    <div style="margin-top:10px; color:var(--muted); font-weight:900; font-size:12px;">
    Projects, tasks and completion are computed from LocalStorage.
    </div>
    </div>
    </div>
    </div>
    </div>
    </div>

<div class="panel mt-4">
<div class="hd">
<h3>Export</h3>
<span>PDF/Excel/CSV (demo)</span>
</div>
<div class="body">
<div class="d-flex flex-wrap gap-2">
    <button class="btn secondary" id="btn_export_json"><i class="fa-solid fa-file-code me-2">
    </i>Export JSON</button>
    <button class="btn secondary" id="btn_export_excel" disabled><i class="fa-solid fa-table-cells me-
    2"></i>Excel</button>
    <button class="btn secondary" id="btn_export_pdf" disabled><i class="fa-regular fa-file-pdf me-
    2"></i>PDF</button>
    </div>
</div>
```

Project Management System

```
<div class="mt-3" style="color:var(--muted); font-weight:900; font-size:12px;">
CSV/JSON export works in this template. PDF/Excel are placeholders for future enhancement.
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
</main>
```

```
</div>
```

```
<script>
```

```
const items = document.querySelectorAll('.sidebar .item[data-page]');
items.forEach(it => it.addEventListener('click', () => location.href = it.getAttribute('data-page')));
const hb = document.getElementById('sidebar_hamburger');
const sb = document.getElementById('app_sidebar');
if(hb && sb) hb.addEventListener('click', () => sb.classList.toggle('open'));
</script>
```

```
<script src="https://cdn.jsdelivr.net/npm/chart.js@4.4.1/dist/chart.umd.min.js"></script>
```

```
<script src="script.js"></script>
```

```
<script>
```

```
function getAll(){
const projects = JSON.parse(localStorage.getItem('pms_projects') || '[]');
const tasks = JSON.parse(localStorage.getItem('pms_tasks') || '[]');
return {projects,tasks};
}
```

```
function buildCharts(){
const {tasks} = getAll();
const completedTasks = tasks.filter(t=>t.status==='Completed').length;
const totalTasks = tasks.length;
const completionRate = totalTasks ? (completedTasks/totalTasks)*100 : 0;
```

```
document.getElementById('success_rate_value').textContent = `${Math.round(completionRate)}%`;
document.getElementById('success_rate_bar').style.width = `${Math.round(completionRate)}%`;
document.getElementById('success_rate_label').textContent = completedTasks + ' of ' + totalTasks + ' tasks completed';
document.getElementById('success_rate_label').textContent = completedTasks + ' of ' + totalTasks + ' tasks completed';
`$ {completedTasks} / $ {totalTasks}` : '—';
```

Project Management System

```
const statusOrder = ['To Do','In Progress','Review','Testing','Completed'];
const statusCounts = statusOrder.map(s=>tasks.filter(t=>t.status===s).length);

const priCounts = {
  Low: tasks.filter(t=>t.priority==='Low').length,
  Medium: tasks.filter(t=>t.priority==='Medium').length,
  High: tasks.filter(t=>t.priority==='High').length,
  Critical: tasks.filter(t=>t.priority==='Critical').length
};

const colors = { blue:'#2563EB', cyan:'#06B6D4', green:'#10B981', amber:'#F59E0B', red:'#EF4444'
};

const isDark = document.documentElement.dataset.theme==='dark';
const gridColor = isDark ? 'rgba(226,232,240,.12)' : 'rgba(2,6,23,.08)';
const tickColor = isDark ? '#F8FAFC' : '#475569';

new Chart(document.getElementById('chart_project_completion'), {
  type:'doughnut',
  data:{
    labels:['Completed','Pending'],
    datasets:[{ data:[completedTasks, Math.max(0,totalTasks-completedTasks)], backgroundColor:
[colors.green, colors.amber], borderWidth:0 }]
  },
  options:{
    responsive:true,
    plugins:{ legend:{ position:'bottom', labels:{ color: tickColor, font:{ weight:'900' } } }, tooltip:{ } },
    cutout:'62%',
    animation:{ duration:900 }
  }
});

new Chart(document.getElementById('chart_task_status'), {
  type:'bar',
  data:{ labels:statusOrder, datasets:[{ data:statusCounts, backgroundColor:[colors.blue, colors.cyan,
colors.cyan, colors.amber, colors.green], borderWidth:0 }] },
  options:{
    responsive:true,
    scales:{ y:{ grid:{ color:gridColor }, ticks:{ color: tickColor, font:{ weight:'800' } } }, x:{ ticks:{
color: tickColor, font:{ weight:'800' } }, grid:{ display:false } } },
  }
});
```

Project Management System

```
plugins:{ legend:{ display:false }, tooltip:{ intersect:false, mode:'index' } },
animation:{ duration:900 }
}
});
```

```
new Chart(document.getElementById('chart_priority'), {
type:'pie',
data:{ labels:['Low','Medium','High','Critical'], datasets:[ { data:[priCounts.Low, priCounts.Medium,
priCounts.High, priCounts.Critical], backgroundColor:['#94A3B8', colors.blue, colors.amber,
colors.red] } ] },
options:{
responsive:true,
plugins:{ legend:{ position:'bottom', labels:{ color: tickColor, font:{ weight:'900' } } } },
animation:{ duration:900 }
}
});
```

```
const labels = ['Jan','Feb','Mar','Apr','May','Jun'];
const values = labels.map((_,i)=> Math.round(completionRate + (i-2)*3));
```

```
new Chart(document.getElementById('chart_monthly'), {
type:'line',
data:{
labels,
datasets:[ { label:'Progress %', data: values.map(v=>Math.max(0,Math.min(100,v))), borderColor:
colors.blue, backgroundColor:'rgba(37,99,235,.15)', fill:true, tension:.35, pointRadius:3 } ]
},
options:{
responsive:true,
scales:{ y:{ grid:{ color:gridColor }, ticks:{ color: tickColor, font:{ weight:'800' } },
suggestedMin:0, suggestedMax:100 }, x:{ ticks:{ color: tickColor, font:{ weight:'800' } }, grid:{
display:false } } },
plugins:{ legend:{ display:false }, tooltip:{ mode:'index', intersect:false } },
animation:{ duration:900 }
}
});
}
```

```
function renderSummary(){
```

Project Management System

```
const {projects,tasks} = getAll();
const completed = tasks.filter(t=>t.status==='Completed').length;
const pending = tasks.length - completed;
const rate = tasks.length ? (completed/tasks.length*100) : 0;

const html = [
  {k:'Projects', v: projects.length, cls:'primary', icon:'fa-solid fa-diagram-project'},
  {k:'Tasks', v: tasks.length, cls:'cyan', icon:'fa-solid fa-list-check'},
  {k:'Completed', v: completed, cls:'green', icon:'fa-solid fa-circle-check'},
  {k:'Pending', v: pending, cls:'amber', icon:'fa-solid fa-clock'}
].map(x=>{
  const pillCls = x.cls==='primary'?'pill primary': x.cls==='cyan'?'pill cyan': x.cls==='green'?'pill green': 'pill amber';
  return `<div class="list-item"><div class="li-left"><div class="ico"><i class="${x.icon}"></i>
</div><div class="txt"><strong>${x.k}</strong><small>${x.v} items</small></div></div><div class="li-right"><span class="${pillCls}">${x.v}</span></div></div>`;
}).join("");

document.getElementById('reports_summary_list').innerHTML = html;

document.getElementById('btn_export_csv')?.addEventListener('click', ()=>{
  const rows = [['type','id','name','status','priority','deadline','projectId'], ...tasks.map(t=>
['task',t.id,t.name,t.status,t.priority,t.deadline,t.projectId])];
  const csv = rows.map(r=> r.map(v=>`"${String(v??'').replace(/"/g,'""')}"`).join(',')).join('\n');
  const blob = new Blob([csv], {type:'text/csv;charset=utf-8;'});
  const url = URL.createObjectURL(blob);
  const a = document.createElement('a');
  a.href=url; a.download='project_management_tasks.csv';
  a.click();
  URL.revokeObjectURL(url);
});

document.getElementById('btn_export_json')?.addEventListener('click', ()=>{
  const payload = {projects,tasks,exportedAt:new Date().toISOString()};
  const blob = new Blob([JSON.stringify(payload,null,2)], {type:'application/json'});
  const url = URL.createObjectURL(blob);
  const a = document.createElement('a');
  a.href=url; a.download='project_management_export.json';
  a.click();
});
```

Project Management System

```
URL.revokeObjectURL(url);
});
}
```

```
document.addEventListener('DOMContentLoaded', ()=>{
renderSummary();
buildCharts();
});
</script>
```

```
<footer style="margin-left:0; padding:18px 22px 24px;">
  <div class="panel" style="border-radius:16px; padding:14px 16px; background:
  rgba(255,255,255,.45);">
<div style="display:flex; flex-wrap:wrap; gap:10px; justify-content:flex-start; align-items:center;
padding-left:50%; transform:translateX(-50%);">
  <div style="color:var(--muted); font-weight:900; font-size:12px;">
Mini Project Developed By:
  <span class="kbd">23AK1A3621 (Janaka Tejeswar)</span>
  <span class="kbd">23AK1A3632 (Rohith)</span>
  <span class="kbd">24AK5A3604 (Nani)</span>
  </div>
  <div style="color:var(--muted); font-weight:900; font-size:12px;">
Note: Still features are under development.
  </div>
</div>
</div>
</div>
</footer>
</body>
</html>
```

Settings Module

The Settings Module allows users to customize application behavior according to their preferences. Configuration options include theme selection, notification management, and interface personalization settings. User preferences are stored within LocalStorage, ensuring that configurations persist across browser sessions.

Through the integration of these modules, ProjectPulse delivers a cohesive and user-friendly project management environment that supports project planning, execution, monitoring, and reporting activities within a modern web-based interface.

CHAPTER-3

SETUP AND DEPLOYMENT GUIDE

3.1 System Setup

3.1.1 System Setup

ProjectPulse is designed as a lightweight client-side web application that does not require backend services, database servers, or additional software dependencies. The application can be executed directly within any modern web browser, making installation and deployment straightforward and platform-independent. Prior to execution, users should ensure that the complete project directory structure is preserved, including all HTML, CSS, JavaScript, image, and configuration files. Since the application relies entirely on client-side technologies, no server configuration or database initialization procedures are necessary.

The minimum requirements for running the application are as follows:

- A modern web browser such as Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari.
- JavaScript enabled within the browser.
- Access to the complete ProjectPulse project directory.
- Sufficient browser storage availability for LocalStorage-based data persistence.
-

Once these requirements are satisfied, the application can be launched by opening the main entry file, namely `index.html`, using a compatible web browser. Upon initialization, the system loads the required resources and automatically creates default data structures within LocalStorage if no existing records are detected.

3.1.2 Local Deployment

The local deployment process is intended for development, testing, demonstration, and academic evaluation purposes. Since ProjectPulse is implemented as a static web application, deployment can be completed without installing web servers or backend frameworks.

The deployment procedure consists of the following steps:

1. Download or extract the ProjectPulse project files into a local directory.
2. Verify that all project resources, including HTML, CSS, JavaScript, image assets, and configuration files, are available within the project structure.
3. Navigate to the project directory.
4. Open the `index.html` file using a modern web browser.
5. Allow the browser to initialize LocalStorage and load the application interface.
6. Begin using the system by creating projects, tasks, and team records.

The application automatically stores all user-generated information in the browser's LocalStorage environment, ensuring data persistence across page refreshes and subsequent sessions.